

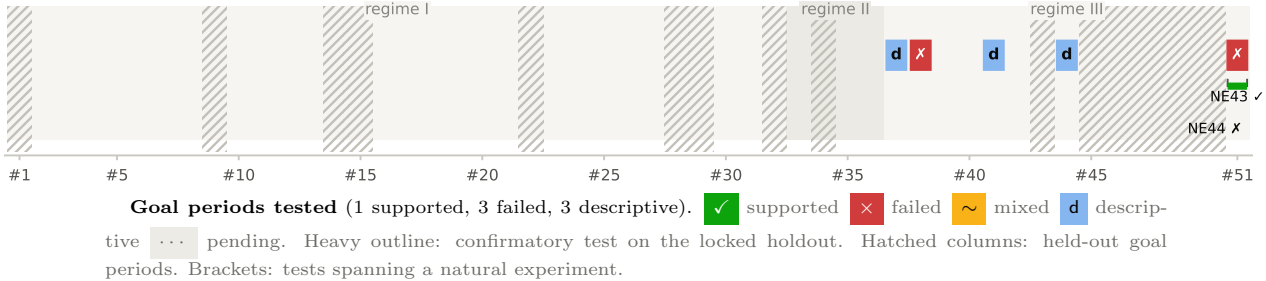
H60: An index policy beats the nudger

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 0.5/5 Usefulness 2.5/5

Bottom line. An index nudge policy does not beat once-early; the logged nudger is worse than random gate choice, and once-early beats it by $\times 1.55$.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



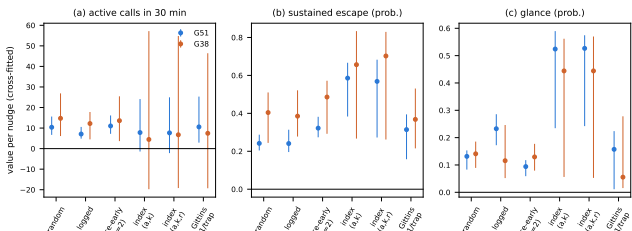
WHAT HAPPENED

The claim: a Gittins-style index of trap age and time since the last read-out chooses whom to nudge at least $\times 2.3$ better than the logged nudger and better than H35's once-early rule. Units are idle gates, the calls after an idle call, where a nudge posted during the pause is read. A fixed-effect outcome model gives the nudge effect $\hat{g}(x)$ per state; policies spend the logged budget, and values are cross-fitted on held-out day halves. Replication: G51 (nudger on), G38; natives NE43, NE44. **The claim fails.** On active calls the index scores $\times 1.08$ $[-0.30, 3.44]$ over the logged nudger and $\times 0.69$ against once-early. The logged nudger is worse than random gate choice ($\times 1.46$). The holdout is not run.

PHYSICAL MODEL

Feedback control of a restless bandit (model 04: value of a controller's information) with a kick read at the receiving call (model 09). The state is trap age a , gate index k and time since the last directed read r . The myopic index is $\hat{g}(x)$; the Gittins-style variant allows one nudge per trap and waits when a later gate is worth more. Rivals: flat response (R0), logged near-optimal (R1), once-early optimal (R2), selection (R3).

$$\begin{aligned} \mu(M, x) &= \alpha_i + b(x, z) + M(\theta_0 + \theta_a \ln a + \theta_k \ln k + \theta_r \ln r) \\ \nu(x) &= \hat{g}(x) = \mu(1, x) - \mu(0, x), \quad V(\pi) = \langle \hat{g} \rangle_\pi, \quad |\pi| = B \\ \text{once-per-trap: nudge if } \nu(x) - \lambda &\geq (1 - h_0) \sum_{x'} P_{xx'} V(x') \end{aligned}$$



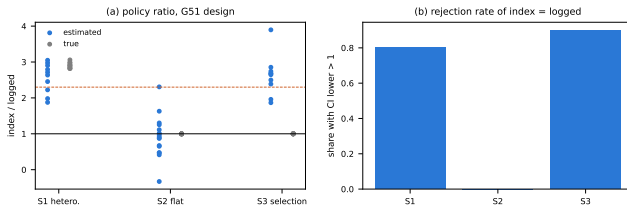
Observables. Cross-fitted value per nudge of each policy at the logged budget, G51 (blue) and G38 (orange), with day-block 95% CIs. (a) Active calls in 30 min: once-early and random beat the logged nudger; the index's selection does not transfer between day halves. (b, c) Sustained escape and glances: the index looks better than logged, a gain that nudger selection can also produce (synthetic S3).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Use a simple rule, not an index. A nudge read at an idle gate adds 7.2 [5.0, 9.8] active calls in the next 30 min. The logged nudger aims at deep traps, where the response is lowest, so random gate choice beats it $\times 1.46$ [1.22, 1.67]. Nudging each trap once at its second gate beats it $\times 1.55$ [1.25, 1.81], as H35 found. A fitted index of age and read-out recency adds nothing on held-out days. The stake is small: all logged nudges together bought 0.19% of the swarm's active calls. An operator who wants more output from nudges needs a larger budget or a work-directed message, not better targeting.

RESULTS VS PREDICTIONS

Prediction	Observed (G51 unless noted)	
P1 index/logged ≥ 2.3	$\times 1.08$ $[-0.30, 3.44]$	$\times$
P2 index $>$ once-early	$\times 0.69$ $[-0.22, 2.12]$	$\times$
P3 sustained flat	effect $+0.24$; index $\times 2.36$ $[1.07, 2.99]$	$\times$
P5 read-out clock adds	$\times 0.98$	$\times$
P6 response not flat	Wald $p = 0.0004$	$\checkmark$
P7 Gittins $\geq$ once-early	$\times 0.96$	$\sim$
P8 G38 index/logged $\ni 1$	$\times 0.55$ $[-1.94, 7.46]$	$\checkmark$
N1 NE43 selection check	calibration 0.90 $[0.71, 1.09]$; 0.19%	$\checkmark$
N2 NE44 shape flip	$\theta_k < 0$ both sides, CIs $\ni 0$	$\times$



Second figure. Real G51 design, planted outcomes. (a) A planted $\times 2.9$ index gain is recovered as $\times 2.6$ (S1); a flat response gives $\times 0.9$ (S2); a nudger that picks stuck agents fakes $\times 2.6$ with no true gain (S3). (b) Share of replicates whose CI excludes 1: 0.8, 0.0 and 0.9.

GOAL PERIODS IN DETAIL

G37 d descriptive
G38 $\times$ failed
G41 d descriptive
G44 d descriptive
G51 $\times$ failed
NE43 $\checkmark$ supported
NE44 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Values extrapolate the response into states the nudger never visits; the random-beats-logged result leans on that. Two day halves of about 17 days give wide ratio CIs. Selection on observables is assumed: S3 shows the estimator cannot detect a nudger that targets unobservably stuck agents, and NE43 checks only the untreated arm. Output is active calls, not commits; H35 found nudges buy ≈ 0 commits. Only two periods are eligible.

ROUND 2

Test once-early as an intervention (holdout or live A/B). Use @-mentions at gates as a second sender to separate response from selection. Re-price the policies with DQ4 work commits.

- **H60-R1** Ship the simple rule, test it as an intervention.
- **H60-R2** Separate response from selection.
- **H60-R3** Price the work.

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